

Subject Description Form

Subject Code	EEE506
Subject Title	Introduction to Wide-bandgap Semiconductors
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims to enable students to master the latest wide bandgap semiconductors materials, properties, and device physics. The subject provides in-depth coverage of wide-bandgap semiconductors materials, <i>i.e.</i> , gallium nitride, silicon carbon, lithium niobite and diamond. It also explores wide-bandgap materials, fabrication, and devices in light, high-frequency communication, high-power electronics applications, etc.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: Category A: Professional/academic knowledge and skills - Understand materials properties: bandgap, optical properties, breakdown field and thermal properties etc. - Understand applications: high power application in large electric fields, light-emitting diodes, transducers and high-electron-mobility transistor etc. - Understand typical wide-bandgap devices design and fabrication: HEMT, Diodes, MOSFET, and LEDs and lasers etc. Category B: Attributes for all-roundedness - Communicate effectively. - Think critically and creatively. - Assimilate new technological development in related field.
Subject Synopsis/ Indicative Syllabus	Introduction overview of latest wide-bandgap semiconductor development, applications, the future, etc. Fundamental properties of wide-bandgap materials: Bandgap, electron mobility, breakdown, thermal conductivity and power handling. SiC materials and devices: Schottky diode and MOSFET; reverse blocking, forward conduction, parasitic capacitance, switching characteristics, on-resistance, gate charge, safe operating area (SOA) and body diodes. SiC and GaN power device fabrication: Three-phase inverter for UPS applications, medium-voltage inverter for PV applications, gate drivers and SiC fabrication process flow. GaN materials and devices: HEMTs, on-resistance, gate threshold voltage, input/output capacitance and fabrication process. GaN Light-emitting Diodes and fabrication: Physics properties and colour, organic light-emitting diodes, and applications. Micro-LEDs, nano-LEDs, high power LEDs and GaN fabrication process. Lithium Niobate devices and fabrication, materials properties and LN waveguide; modulator and switches, LN fabrication and design rule.

Teaching/Learning Methodology	Lectures, supplemented with interactive questions and answers, and short quizzes	a, b, c, d, e, f	In lectures, students are introduced to the knowledge of the subject, and comprehension is strengthened with interactive Q&A and short quizzes					
	Tutorials, where problems are discussed and are given to students for them to solve	a, b, c, d, e, f	In tutorials, students apply what they have learnt in solving the problems given by the tutor.					
	Assignments	a, b, c, d, e, f	Through working assignment, students will develop a firm understanding and comprehension of the knowledge taught.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					
			a	b	c	d	e	f
	1. Assignments	40%	✓	✓	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓	✓	✓
	Total	100 %						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The assignments will assess the students’ understanding of the basic concepts and principles in third-generation semiconductors. The examination will be conducted to make a comprehensive assessment of students’ intended learning outcomes as stated above.							
Student Study Effort Expected	Class contact:							
	▪ Lecture						30 Hrs.	
	▪ Tutorial						9 Hrs.	
	Other student study effort:							
	▪ Lectures/Tutorials, self-study						81 Hrs.	
	Total student study effort						120 Hrs.	

Reading List and References	Textbooks • John H. Lan, Semiconductor Advanced Packaging, Springer, 2021. • Rao R. Tummala, Fundamentals of Microsystems Packaging, McGraw-Hill, 2001. References • Daniel Lu and C. P. Wong, Materials for Advanced Packaging, Springer, 2018. • B. Q. Wu, A. Kumar, S. Ramaswami, 3D IC Stacking Technology, McGraw-Hill, 2011.
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